2. SUBDIVISIONS OF THE CHAIN OF TRIANGULATION AND THEIR DISTINGUISHING CHARACTERISTICS.

The contrast in the physical features along the arc of the thirty-ninth parallel is so well pronounced as conveniently to mark out for general description three subdivisions, which moreover demand, in part at least, different mathematical treatment in the reduction of the observations. These subdivisions are designated the western, the central, and the eastern sections.

The *western* section is characterized by the great altitudes of its stations and the unusually large size of its triangles, many of the sides being over 160 kilometres, or 100 statute miles in length. The triangulation crosses the Coast Range, the Sierra Nevada, the Wasatch Range, and the main ridge of the Rocky Mountains, with many of its stations more than 3 kilometres, or nearly 2 statute miles, above the level of the ocean. The total linear development between Point Arena on the coast and Big Springs off the eastern flank of Pikes Peak, Colorado (as projected on the parallel of 39°) is nearly 1 685 kilometres, or 1 047 statute miles.

The *central* section, which extends from Big Springs, Colorado, eastward as far as St. Louis, Missouri, over a distance of about 1 217 kilometres, or 756 statute miles (measured on the parallel of 39°), partakes of the very opposite character from its neighbor with respect to width of development or average length of sides. The latter is but 27·3 kilometres, or 17·0 statute miles, and is therefore a minimum value. This feature was imposed upon it by the general flatness of the great plain which lies between the eastern slope of the Rocky Mountains and the Mississippi River, descending very gradually from about 1 800 metres (5 900 feet nearly) to about 135 metres, or 443 feet, above the sea level. As a rule the theodolite was mounted on tripods or scaffolds in order to overcome the earth's curvature and keep the line of sight sufficiently elevated above the ground.

The third or *eastern* section differs from the others by its small but diversified hypsometric features being composed of plains, low hills, and mountain ranges. Where the triangulation traverses the Alleghenies altitudes exceeding 1 300 metres, or 4 265 feet, are met. The section crosses the Chesapeake and Delaware bays, terminating at the capes of the latter. Its total (referred) length is about 1 323 kilometres, or 822 statute miles.

The triangulation across the country possesses great internal rigidity by reason of its composition throughout. Either quadrilaterals or central figures such as polygons formed by combination of triangles compose the scheme, while its length is supported by 10 base lines suitably distributed.

3. GENERAL STATEMENT IN REGARD TO THE ASTRONOMIC MEASURES.

Respecting the astronomic measures there are 109 stations directly connected with the triangulation at which the latitudes were determined almost *exclusively by Talcott's* method. These observations fall between the years 1846 and 1898. Eight latitudes depend on other than Coast and Geodetic Survey authority. Astronomical azimuths were obtained at 73 of the trigonometric stations between the years 1849 and 1897. A variety of methods, suitable to the circumstances at the time, were employed in this

(G.) THE COMPUTATION OF THE SPHERICAL EXCESS OF THE TRIANGLES.

For all that part of the triangulation which lies east of the Rocky Mountains, and which traverses the plains and gentle slopes of Kansas, Missouri, and Ohio, the comparative shortness of the sides of the triangles admits of the application of Legendre's theorem in its simple form. The spherical excess ε (in seconds) is given by $\frac{a_1 b_1 \sin C_1}{2r^2 \sin 1''}$, where $a_1 b_1 C_1$ refer to sides and included angle of a *plane* triangle, whose angles are those of the corresponding small spherical triangle after each has been diminished by $\frac{1}{3}\varepsilon$. When greater precision is required as for the larger triangles which stretch across the peaks and ridges of the Allegheny Range, we introduce the radius of an osculating sphere (referring to the center of the triangle) and take

$$\varepsilon = \frac{a_1 b_1 \sin C_1}{2\rho_m \rho_n \sin 1''} = \frac{a_1 b_1 \sin C_1}{2a^2 (1 - e^2) \sin 1''} \left[1 - e^2 \sin^2 \phi\right]^2$$

The quantity $\frac{[1 - e^2 \sin^2 \varphi]^2}{2a^2 (1 - e^2) \sin 1''}$ has been tabulated with the latitude φ as argument, for which see Coast and Geodetic Survey annual report for 1894.*

For triangles of unusually large size and approaching the limit for possible observation, certain terms in the development of the theorem which ordinarily could be neglected need examination. It has been shown that spheroidal triangles may be computed as spherical and hence as plane ones by application of the same theorem extended.† Various forms have been given to the development of the theorem. ‡ Let

S_1 = surface of the corresponding plane triangle = $\frac{1}{2} a_1 b_1 \sin C_1$,
and let $m^2 = \frac{1}{3}(a_1^2 + b_1^2 + c_1^2)$, then

$$\varepsilon'' = \frac{S_1}{\rho_m \rho_n \sin 1''} \left(1 + \frac{m^2}{8\rho_m \rho_n} + \ldots\right)$$

where ρ_m and ρ_n are the radii of curvature in the plane of the meridian and normal to it, and ε is to be distributed unequally over the angles,§ viz:

$$A - A_1 = \frac{\varepsilon}{3} + \frac{\varepsilon}{60} \cdot \frac{m^2 - a_1^2}{\rho_m \rho_n} + \ldots \text{ or } \frac{\varepsilon}{3}\left(1 + \frac{m^2 - a_1^2}{20\rho_m \rho_n}\right)$$

$$B - B_1 = \frac{\varepsilon}{3} + \frac{\varepsilon}{60} \frac{m^2 - b_1^2}{\rho_m \rho_n} + \ldots \quad \frac{\varepsilon}{3}\left(1 + \frac{m^2 - b_1^2}{20\rho_m \rho_n}\right)$$

$$C - C_1 = \frac{\varepsilon}{3} + \frac{\varepsilon}{60} \frac{m^2 - c_1^2}{\rho_m \rho_n} + \ldots \quad \frac{\varepsilon}{3}\left(1 + \frac{m^2 - c_1^2}{20\rho_m \rho_n}\right)$$

A convenient logarithmic formula has been given by the late C. H. Kummell, tables of the factors log A and log B of the Coast and Geodetic Survey method for the

*Appendix No. 9, pp. 290–291.

†The spherical excess of a spheroidal triangle is equal to that of a spherical triangle whose angular points have the same latitudes and longitudes as the corresponding points of the spheroidal triangle—Clarke's Geodesy (1880), pp. 49 and 107.

‡Helmert's Theorieen der Höheren Geodäsie (1880), vol. 1, pp. 88–101.

§Helmert, ibid, p. 98.

Respecting weights to the several directions entering into the adjustment, it has been decided to give them all the same—that is, unit weight. This proceeding is justified by the following considerations.

From the approximate probable errors of directions in the preceding abstracts we find the average value from 82 directions $= \pm 0''.088$; on the other hand, we derive from the base figure adjustment, as given in the following pages, the probable error of a direction by using $\frac{2}{3}$ of the value resulting from the 56 direction corrections, viz: $\pm 0''.31$, or 3.4 times the first value. We may also use the 99 angular corrections of the 33 triangles, whence we get for the probable error of a direction $\frac{2}{3}\sqrt{\frac{25.0}{2 \times 99}} = \pm 0''.24$. Again, if we operate with the closing errors of the triangles, we find for the probable error of a direction—

$$\frac{2}{3}\sqrt{\frac{37.96}{33 \times 6}} = \pm 0''.30$$

We thus find the probable error of a direction e_t as derived from the corrections demanded by the adjustment of the base net $= \pm 0''.28$ and the same e_s as derived from the station adjustment $= \pm 0''.09$; the fact that e_t is three times as great as e_s is attributed mainly to the effect of local deflections in measuring angles, the vertical axis of the theodolite being necessarily adjusted to the plumb-line. Besides, a very careful adjustment of the instrument is required when the station observed upon is considerably above or below the one occupied. Persistent lateral refraction also has a share in producing the above result. Following the methods outlined at the beginning of this paper and used in the adjustment of the Yolo Base Net, we have $e_c = \sqrt{(0.28)^2 - (0.09)^2} = \pm 0.27$, which is to be combined with the particular value of e_s; hence the relative weight of an observed direction becomes—

$$p = \frac{1}{e^2} = \frac{1}{e_s^2 + (0.27)^2}$$

In the case of the Salt Lake Base Net, we have in the main figure the *extreme* values of $e_s \pm 0''.06$ and $\pm 0''.15$; hence the extreme weights to directions would be in the proportion of 13 to 11 nearly. The introduction of weights was therefore deemed unnecessary, especially when we consider the strength of the development of the length of the base to that of the primary line.

Designation of triangulation.	Starting and junction lines.	Distances between sides of base nets.	Number of intermediate triangulation stations.	Number of conditional equations involved.
		km.		
1 The Eastern Shore series	Cape May Light to Cape Henlopen Light Finlay to Linstid	140	14	18+15
2 The Allegheny series	Webb to Marriott Summersville to Ivy	445	20	22+33
3 The Ohio series	Piney to Pigeon Reizin to Culbertson	280	23	50
4 The Indiana series	Green to Stout Hunt City to Claremont	216	15	34
5 The Illinois series	Hunt to Newton Clark Mound to Dreyer	171	12	33
6 The Missouri series	Insane Asylum to Kleinschmidt Christian to Belche	192	22	65
7 The Missouri and Kansas series	Hubbard to Hughes Vine Creek to Iron Mound	402	36	77
8 The Kansas and Colorado series	Thompson to Heath Holcomb Hill to Big Springs	560	42	99
9 The Rocky Mountains series	Divide to Big Springs Mount Nebo to Ibepah	780	12	28
10 The Nevada and California series	Mount Nebo to Ibepah Mount Helena to Mount Diablo	850	16	40
11 The Western or Coast Range series	Mount Helena to Mount Diablo Fisher to Cold Spring	160	8	23

The total number of principal triangulation stations, not counting those of the base nets, is 220. Adding to these the latter, or 88, we have for the total number of principal stations 308. To these must be added about 240 subordinate stations—i. e., those which connect the geodetic and astronomic positions.

The total number of conditions in the above series is 537. Adding to these the 206 conditions in the base nets, the grand total of conditions subsisting and satisfied is 743.

The following tables contain statistics relating to the angular measures, the closure of triangles, and the degree of accord between any two adjacent base lines when connected by a series of triangles.

With respect to the closure of the triangles (π plus spherical excess minus the sum of the angles), we find the number of cases in excess to those in defect in the ratio 36 to 34, nearly. If Δ equal the closing error, and n the number of triangles, the column headed "Mean error of an angle" gives the quantity $a = \sqrt{\frac{[\Delta^2]}{3n}}$ with an average value of $\pm 0''.77$. The column headed "Probable error of an adjusted direction" is given by $d = 0.675 \sqrt{\frac{[vv]}{c}}$ where c = number of conditions.

The average value is $\pm 0''.44$

H. SUMMARY OF RESULTS RELATING TO ANGULAR MEASURES.

Designation of locality.	Triangle closing errors. Number of +	Number of −	Sum of +	Sum of −	Number of triangles.	Mean error of an angle.	Probable error of a resulting direction.
			"	"		"	"
Eastern Shore series	14	14	26·55	22·62	28	±1·22	±0·72
Kent Island Base Net	6	6	8·42	7·54	12	0·96	0·41
Allegheny series	33	* 18	46·79	24·20	52	0·98	0·45
St. Albans Base Net	11	14	11·35	23·15	25	1·04	0·47
Ohio series	20	24	24·53	25·77	44	0·85	0·45
Holton Base Net	4	11	4·40	7·47	15	0·58	0·34
Indiana series	15	12	13·60	8·11	27	0·60	0·34
Olney Base Net	13	22	11·01	16·18	35	0·54	0·29
Illinois series	10	18	6·85	15·37	28	0·57	0·34
American Bottom Base Net	7	9	11·31	24·17	16	1·59	0·82
Missouri series	36	28	49·72	19·95	64	0·81	0·66
Versailles Base Net	17	14	16·00	11·83	31	0·64	0·40
Missouri-Kansas series	41	29	36·40	24·90	70	0·60	0·35
Salina Base Net	8	6	8·73	7·10	14	0·75	0·44
Kansas-Colorado series	48	48	50·24	46·05	96	0·75	0·50
El Paso Base Net	7	9	4·94	10·11	16	0·68	0·40
Rocky Mountain series	13	10	11·29	7·93	23	0·57	0·32
Salt Lake Base Net	18	15	14·43	12·28	33	0·66	0·32
Nevada-California series	15	15	8·00	9·17	30	0·42	0·23
Yolo Base Net	7	12	2·09	10·79	19	0·51	0·24
Western or Coast Range series	14	9	16·56	24·91	23	±1·37	±0·67
Sums	357	343	383·21	359·60	701		
Average value from 701 triangles						±0·77	
Average value from 1 660 directions							±0·44

*One triangle closes exactly.

I. ACCORD OF THE BASE LINES.

In the adjustment of the triangulation between two adjacent base nets the length equation has been derived from the angles as given by the station adjustments previous to any further adjustment, the triangles not even having been closed. Any route might have been selected, but such angles as differ least from 90° have been chosen. Spherical angles have been used, the logarithms of the terminal lines having been corrected for difference in arc and sine. In the solution of the normal equations the length equation was assigned the last place in order, so that the discrepancy was corrected for the adjustment of all the other equations, thus showing the final discrepancy which was distributed over the figure, and which was the same that would have been obtained if the length equation had not been formed until all the other equations had been adjusted.

The following table shows the discord in length between adjacent base lines as computed through the intervening triangulation, derived in the manner explained above. A plus sign indicates that the base to the east gives the greater length. The discrepancy is given in units of the seventh decimal place of logarithms and also in parts of the length.

Base lines.	Discrepancy.	
	In logarithm.	One part in—
Kent Island and St. Albans	+ 11	395 000
St. Albans and Holton	— 24	181 000
Holton and Olney	— 71	61 200
Olney and American Bottom	— 6	724 000
American Bottom and Versailles	+ 86	50 500
Versailles and Salina	+169	25 700
Salina and El Paso	— 92·3	47 000
El Paso and Salt Lake	+ 85·4	50 840
Salt Lake and Yolo	+ 82·6	52 600